Background

I first met Gordon Pask in about 1963. I worked for him for about a year in a commercial company he had founded to exploit the results of some of his research – especially keyboard training machines, but also other electronic devices. Most of my work there was on developing a prototype electronic desk calculator for the Swedish company Facit – at this time almost all calculators were mechanical (see http://home.vicnet.net.au/~wolff/calculators/facit/facit.htm). Our earliest working version occupied a six foot rack, (that is, it was about 2m tall by 1m wide by 50cm deep). We were working to miniaturize it to desktop size when Facit gave up and stopped funding. After working a couple of years in other electronics companies, I returned to work for Pask (in about 1966) at his research company, System Research Limited, developing electronics for various experimental and research projects.

You need to remember that at this time computers were almost all large and expensive, and a small poorly funded research organization like Systems Research couldn't afford one. Pask's especial genius was in devising technologically simple and cheap ways to explore conceptually complex issues and problems. Tony Watts had been working for Pask for a long time, building experimental equipment using electro-mechanical relays. I added skills in electronics, using individual components (transistors etc) on circuit boards to implement more complex functions (integrated circuits were first developed in 1962, and were still expensive and not widely available). This was the technology (electro-mechanical relays and simple electronics using discrete components) used to build the colloquy of mobiles.

As you know, available technology evolved fast in subsequent years. By 1968 Peter Zinovieff, the electronic music composer for whom I had also built equipment had acquired his first computer (a Digital Equipment Corporation PDP-8 which was included in his exhibit for Cybernetic Serendipity – the only computer in the exhibition), but it was a few more years before individual ownership of computers would become relatively common.

Zinovieff's exhibit allowed visitors to whistle or sing a tune into a microphone, and the computer would then generate a set of variations on the theme provided. As I recall, at the official exhibition opening, Princess Margaret whistled the British National Anthem.

Colloquy of Mobiles

The female figures were designed by Yolanda Sonnabend, the stage designer (use Google for lots of information about her). They were built (by Pip and Adele Youngman) of fiberglass, with each body cast in two pieces, plus a separate "head" casting, and about 1.8m – 2m tall. The figures were off-white, but internally illuminated by coloured light bulbs (whose intensity could change) giving the impression that they were of (changeable) colours. They were suspended from the ceiling by a shaft driven by a motor that could make them rotate (clockwise/anticlockwise by about 180 degrees). One clockwise/anticlockwise rotation would take about 30 seconds. Internally, each female had a mirror (visible through a hole in the middle of the body, that could rotate on a horizontal axis by ~180 degrees, with one up/down rotation about every 10 seconds. Inside each female body was a loudspeaker, that could make a "hooting" noise (a bit like an automobile horn), and a microphone to detect male sounds.

Each male body, constructed of aluminium, was about 1m X 0.5mX10cm. They were mostly black. Like the females, the males were suspended, and could be rotated by a motor. In the center of each male was an automobile headlight (positioned at the same level as the female mirrors). Males also had a microphone, to detect female sounds, a loudspeaker, to make a male "hooting" sound, and photocells dangling above and below the body to detect light reflected by the females (they look like small Calder mobiles in the photo). The males (I think) also had signal lamps on their surfaces that indicated the level of their "drive" (see below).

There was no control electronics inside any of the mobiles. Wiring from each mobile led through the ceiling to a six foot rack containing the electronics.

I'm not sure (can't remember) whether the control electronics exactly implemented the flow charts shown in Pask's diagrams. It is quite likely that the actual implementation was slightly simpler than that shown in the flow charts. For example, the flow charts show that there were two possible male lamp colours (“A” and “B”, or Orange and Puce). The main light on the male mobiles was actually a single colour (white light), so I suspect that only about half of each flow chart was actually built (technically, it would have been difficult to distinguish between two different light colours in a visually noisy environment). This simplification wouldn’t really make much difference to the complexity of interaction between the mobiles, or to the conceptual point that Pask was trying to make.

The best way to think of the interactions between the mobiles is in sexual terms – this was certainly how Pask thought and talked about it even if he didn’t make it explicit in what he wrote (where he talks about “satisfaction”). Each mobile had some level of “sex-drive” that increased over time (and, for the females, was signaled by an increasing level of internal illumination). When this reached some threshold, the mobile initiated activity to try to have sex, which could only be achieved by a male shining a light at the mirror in the middle of a female, and having the light reflected back at the dangling photocells above or below the male body. This resulted in “orgasm”: the female would stop rotating, stop swiveling the mirror, and make a hooting sound (detected by a microphone in the male, which would also stop rotating. Orgasm would quickly reduce the sex drive of the two participants (the female light would dim), and both would go into a quiescent state until their sex drives built up again.

It was certainly possible for people to participate in the interactions between the mobiles (flashlights were provided), and some did. Shining a light at the photocells of a receptive male would trick it into thinking you were a female, but it was harder to trick the females into orgasm, because you had to make exactly the right noise at the right time. I don’t think that many of the audience who interacted understood very clearly what was going on – to interact effectively, and fully participate in the interaction, you needed to have a clear mental model of how the mobiles were interacting with each other, and this was hard to get from observation, or from reading Pask’s description.